Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

1.(a)

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
   "Year":[2015,2017,2019,2021,2023,2024],
   "Jobs":[1000,1800,3000,5000,7500,9000]
})

plt.plot(df["Year"], df["Jobs"], marker="o")
plt.xlabel("Year")
plt.ylabel("Jobs")
plt.title("Data Science Job Trend")
plt.show()
```

Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

1.(b)

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.DataFrame({
   "Role":["Analyst","Engineer","Scientist","ML Engineer"],
   "Jobs":[3000,2500,2000,1500]
})

plt.bar(df["Role"], df["Jobs"])
plt.xlabel("Role")
plt.ylabel("Jobs")
plt.title("Data Science Roles")
plt.show()
```

Conduct an experiment to differentiate Structured, Un-structured and Semi structured data based on data sets given.

1.(c)

Code:

```
import pandas as pd

# Structured
df = pd.DataFrame({"Name":["A","B"],"Age":[20,21]})
print("Structured:\n", df)

# Semi-structured
data = {"Name":"A","Age":20}
print("Semi-structured:\n", data)

# Unstructured
text = "A is a student interested in Data Science."
```

```
print("Unstructured:\n", text)
```

Conduct an experiment to encrypt and decrypt given sensitive data. Use the cryptography library fernet.

```
from cryptography.fernet import Fernet

# Generate key
key = Fernet.generate_key()
f = Fernet(key)

# Original data
data = "Hello123".encode()

# Encrypt
encrypted = f.encrypt(data)
print("Encrypted:", encrypted)

# Decrypt
decrypted = f.decrypt(encrypted)
print("Decrypted:", decrypted.decode())
```